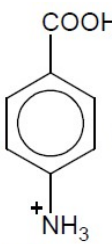


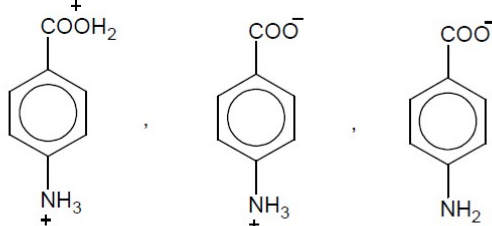
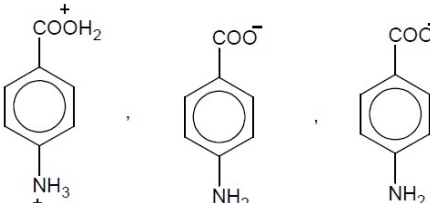
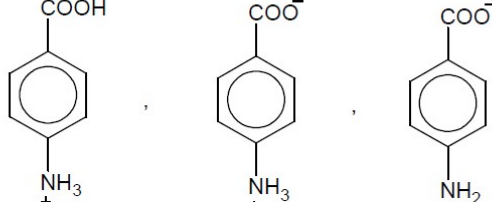
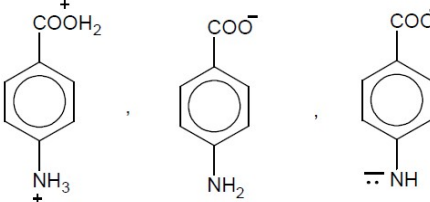
PHYSICS PTA-1 (ASSIGNMENT)

PT-1

SINGLE ANSWER TYPE:

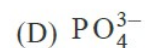
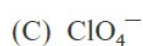
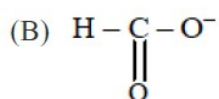
1.

Major species obtained when  is dissolved in H₂O at pH = 2, pH = 7 and pH = 12, respectively are? $pK_{a_1} = 2.35, pK_{a_2} = 4.98$.

- (A)  , (C) 
- (B)  , (D) 

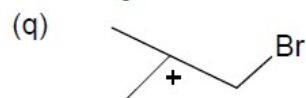
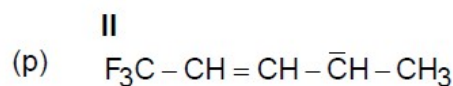
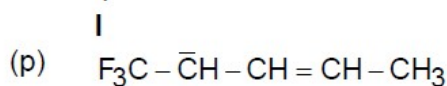
2.

Which of the following oxyanion has 3 identical resonating structures?



3.

For the pairs of canonical forms written below what is/are true?



- (A) In both (p) and (q), I is more stable than II
(B) In both (p) and (q), II is more stable than I
(C) In (q), II is more stable than I
(D) In (p), II is more stable than I

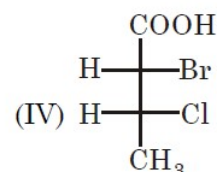
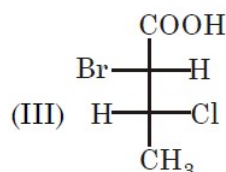
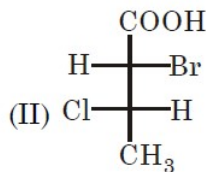
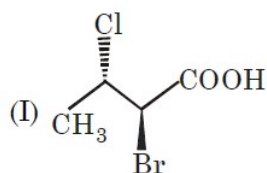
4.

(+)-2-butanol has $[\theta]_D^{25} = 13.9^\circ$. A sample of 2-butanol containing both the enantiomers was found to have specific rotation value of -3.5° under similar condition. The percentage of the (+) and (–) enantiomer present in the sample are, respectively :

- (A) 37.4% and 62.6% (B) 62.6% and 37.4%
(C) 42.2% and 57.8% (D) 35.5% and 64.5%

5.

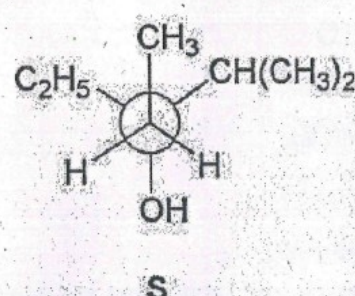
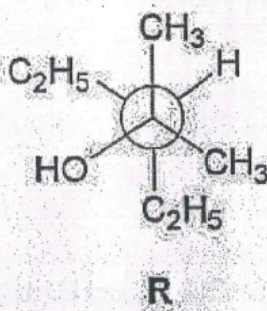
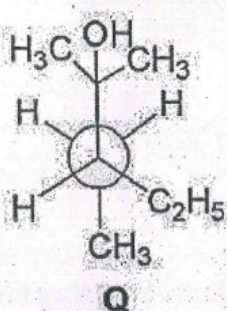
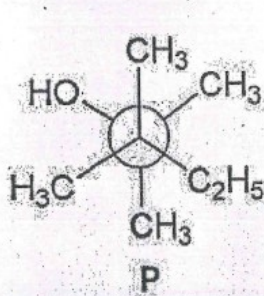
Identify the pair of compounds having same solubility in H_2O



- (A) I and II (B) I and III (C) II and IV (D) I and IV

6.

Newman projections P, Q, R and S are shown below :



Which one of the following options represents identical molecules ?

- (A) P and Q (B) Q and S (C) Q and R (D) R and S

7.

	Column-I		Column-II
(P)		1.	Diastereomer
(Q)		2.	Identical
(R)		3.	Enantiomer
(S)			

(A) P → 2, Q → 3, R → 2, S → 2

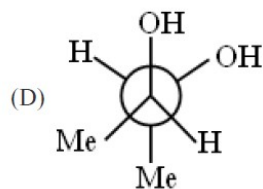
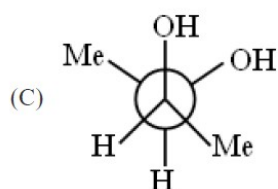
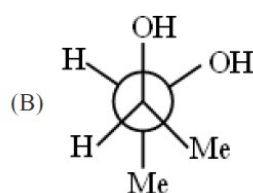
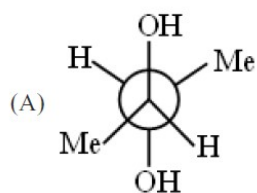
(B) P → 3, Q → 1, R → 1, S → 2

(C) P → 2, Q → 1, R → 1, S → 3

(D) P → 2, Q → 3, R → 3, S → 1

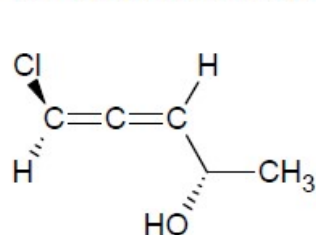
8.

Among the following, the conformation that corresponds to the most stable conformation of *meso*-butane-2,3-diol is –

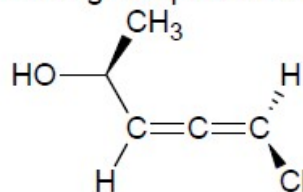


9.

The correct relation between the following compounds is:



and



(A) Enantiomers

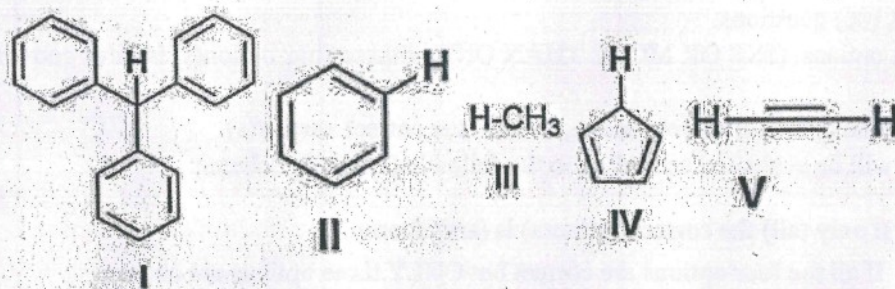
(B) Diastereomers

(C) Homomers (identical)

(D) Constitutional isomers

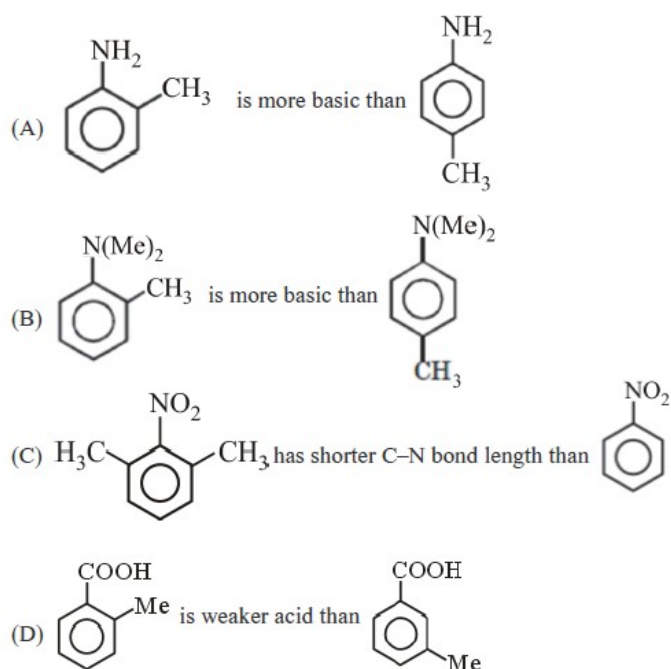
MULTI SINGLE ANSWER TYPE:

10. With respect to the compounds I-V, choose the correct statement(s).

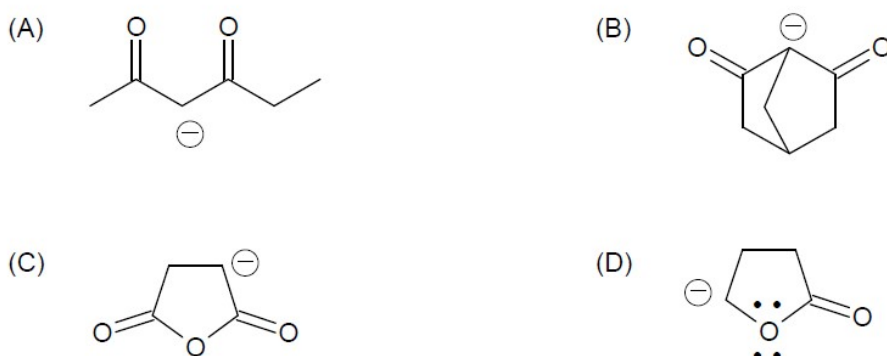


- (A) The acidity of compound I is due to delocalization in the conjugate base.
 (B) The conjugate base of compound IV is aromatic.
 (C) Compound II becomes more acidic, when it has a $-\text{NO}_2$ substituent.
 (D) The acidity of compounds follows the order $\text{I} > \text{IV} > \text{V} > \text{II} > \text{III}$.

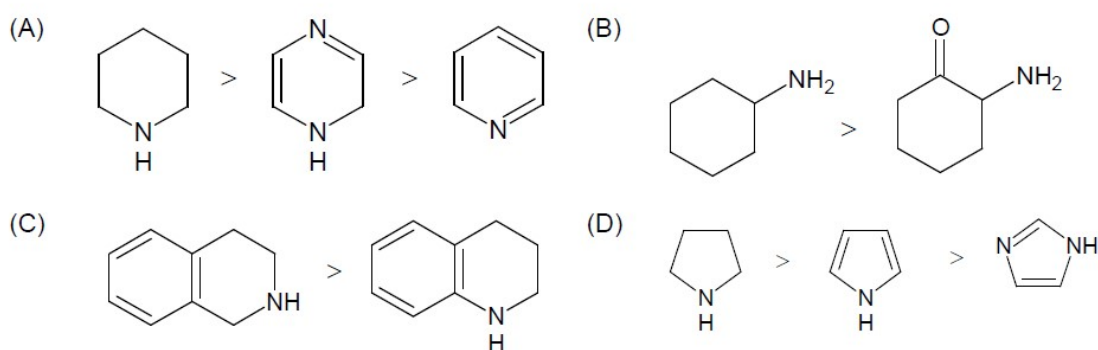
11. Select the incorrect statement(s) ?



12. Which of the following carbanions are not resonance stabilized?



13. Which of the following order for basic strength is/are correct?



14. Select incorrect statements.

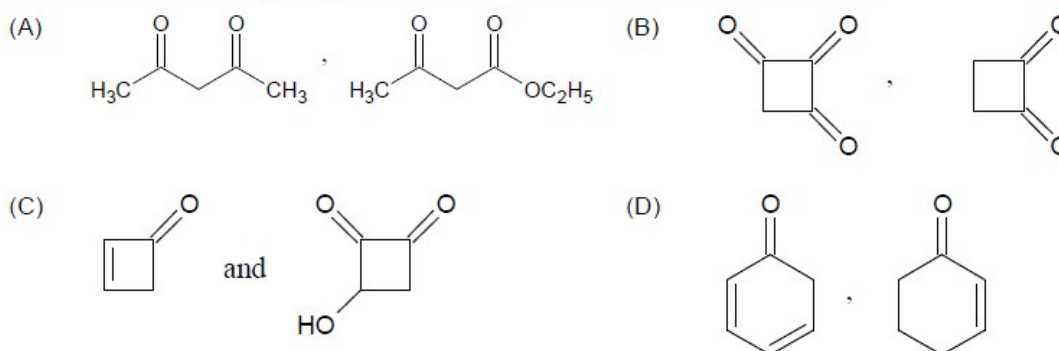
(A) Conjugate acid of pyrrole is anti-aromatic.

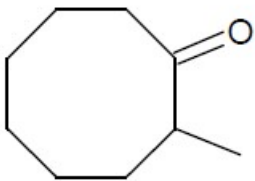
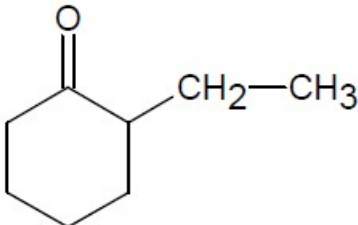
(B) $\text{Me}_3\text{C}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{ND}_2$ can't tautomerise

(C) CH_3-S^- is less stable than CH_3-O^-

(D) Optically pure $\text{EtCH}(\text{Me})-\overset{\text{Me}}{\underset{\text{H}}{\text{C}}}-\text{CH}=\text{O}$ racemize during tautomerisation.

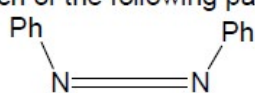
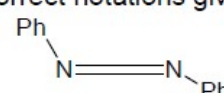
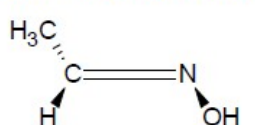
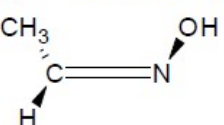
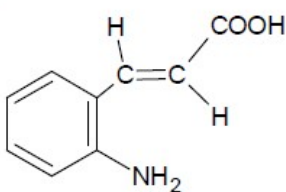
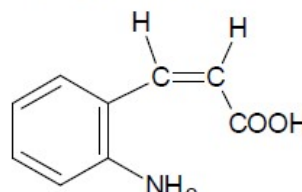
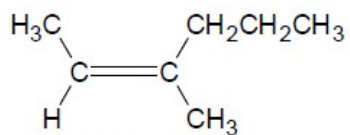
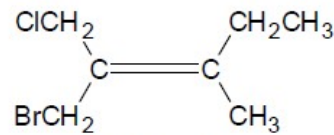
15. In which of the following compounds, second has lesser enol content than first:



16. Ethyl acetoacetate gives the test of:
- (A) $\begin{array}{c} \diagup \\ \text{C}=\text{O} \\ \diagdown \end{array}$ and $\begin{array}{c} \text{O} \\ \parallel \\ \text{—C—OC}_2\text{H}_5 \end{array}$ groups
- (B) $\text{CH}=\text{CH}$ and OH groups
- (C) $\begin{array}{c} \text{O} \\ \parallel \\ \text{—C—O—C}_2\text{H}_5 \end{array}$ and OH group
- (D) $-\text{CHO}$ group
17. Tautomer of which of the following can show geometrical isomerism
- (A) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{H}$
- (B) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{H}$
- (C) 
- (D) 

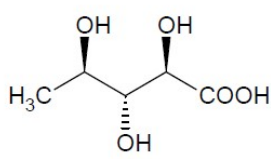
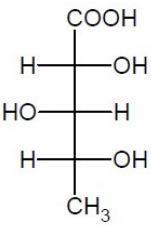
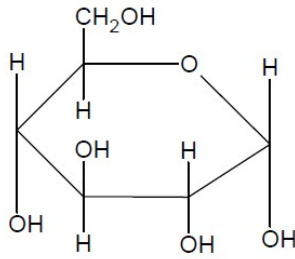
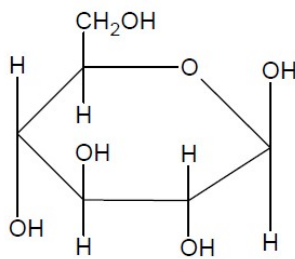
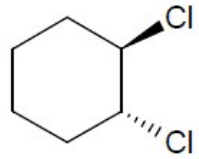
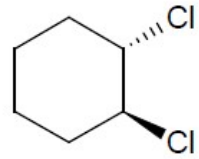
18.

For which of the following pairs of compounds are the correct notations given?

- (A)  and 
Anti - azobenzene Syn - azobenzene
- (B)  and 
Syn - acetaldoxime Anti - acetaldoxime
- (C)  and 
Trans - O - aminocinnamic acid Cis - O - aminocinnamic acid
- (D)  and 
Z - isomer E - isomer

19.

Which of the following pair of compounds does not represent enantiomeric pair?

- (A)  and 
and
- (B)  and 
and
- (C)  and 
and
- (D)  and 
and

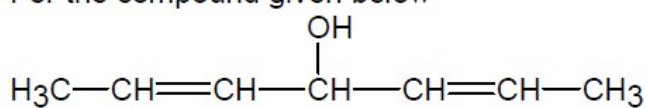
20. Identify the correct option(s)
- (A) Diastereomers can be separated by fractional distillation
 - (B) Enantiomers are also called as optical antipodes
 - (C) 'Epimers' and 'Anomer's are the type of diastereomers
 - (D) Geometrical isomers are diastereomers

PASSAGE TYPE:

If 'x' = Number of diastereomeric pair and

'y' = Number of optical active isomers

For the compound given below



Then

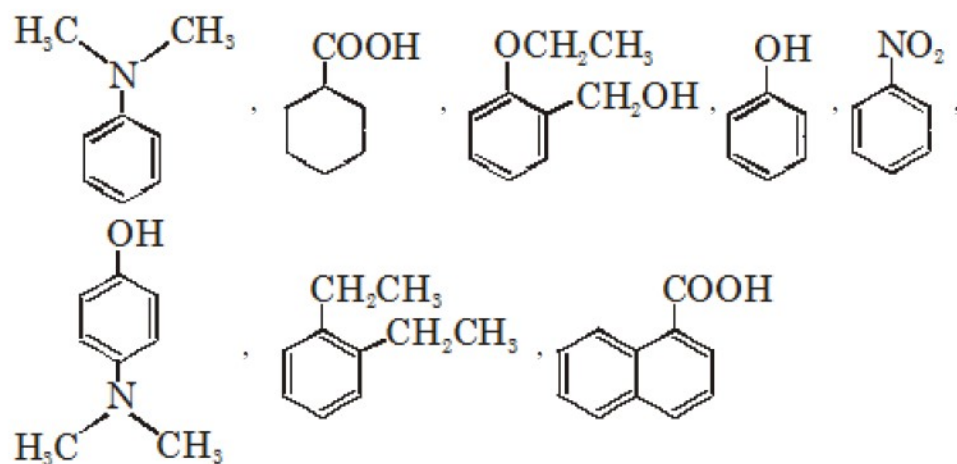
21. The value of x is _____

22. The value of y is _____

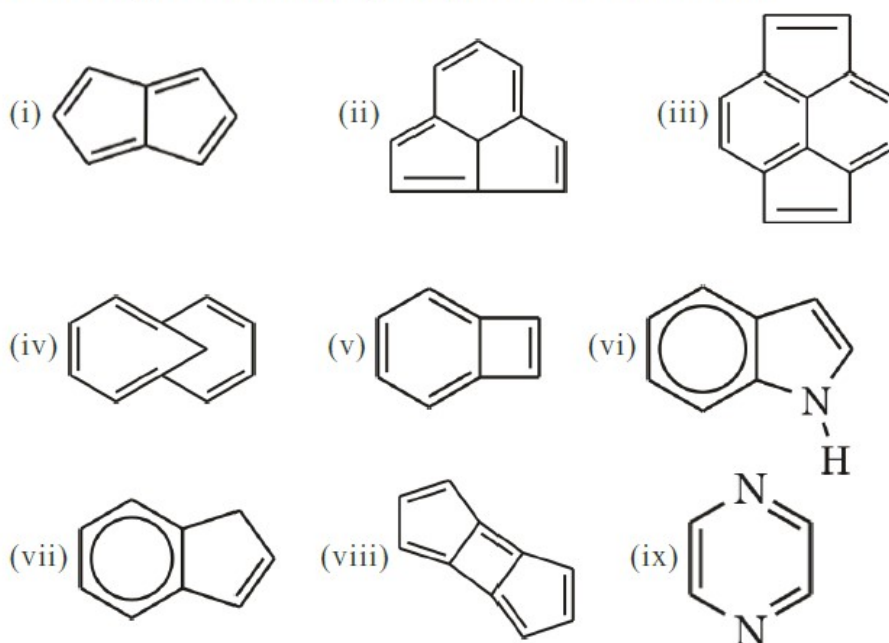


INTEGER TYPE:

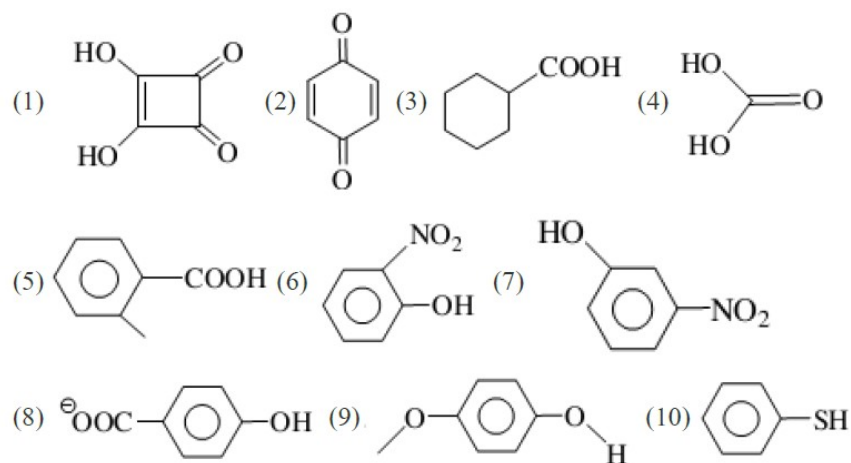
23. Amongst the following, the total number of compounds soluble in aqueous NaHCO_3 is:



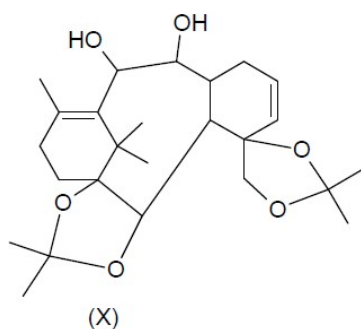
24. How many of the following compounds are not aromatic.



25. Compound having less pH value than phenol.



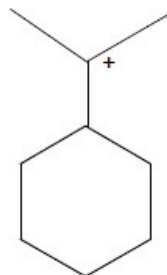
26.



Degree of unsaturation in 'X' is equal to a; and if $b = \frac{a \times 10}{2}$. Then what is the value of b?

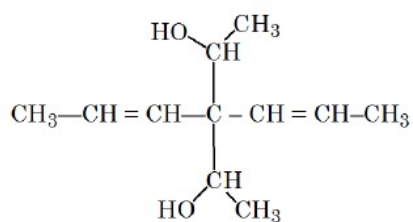
27.

What is the number of hyperconjugative structures of the given carbocation



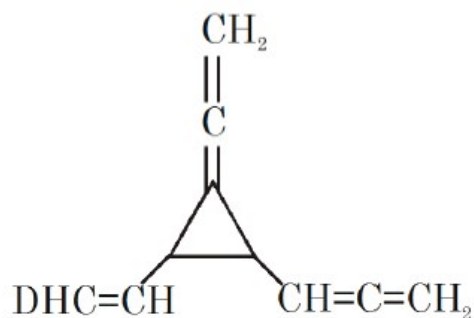
28.

Total number of stereoisomers for following structure is 'X' Then identify the value of (X - 5) is :-

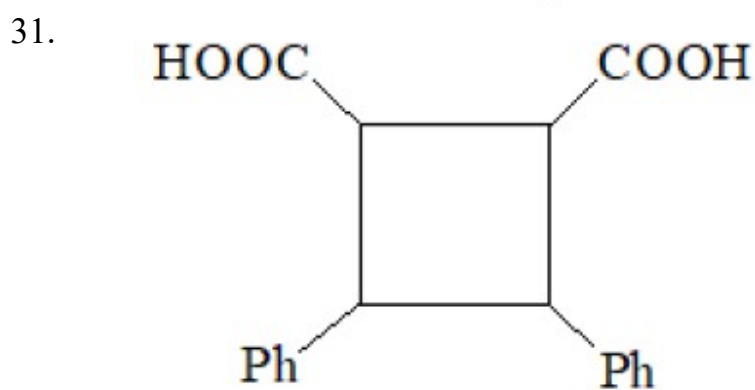
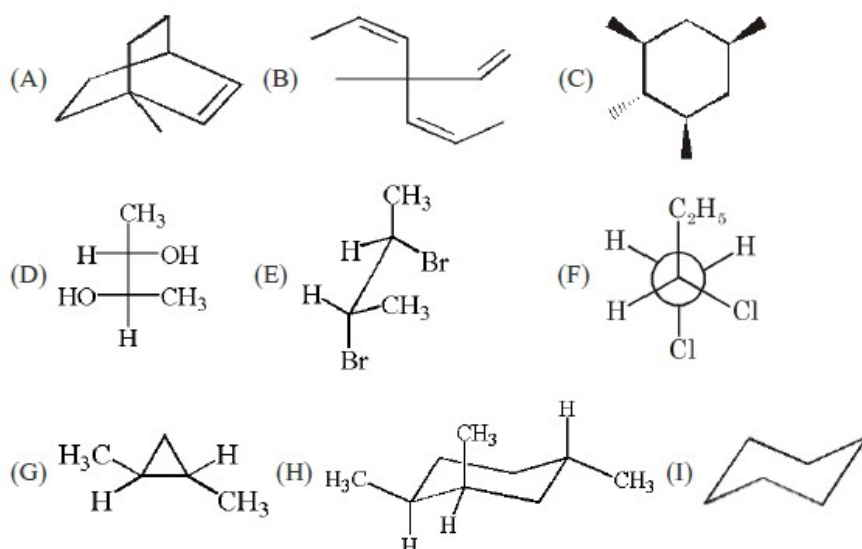


29.

Total number of stereo isomers corresponding to structure



30. Which of the following is/are chiral compounds?



Number of optically active isomers = x

Number of meso compounds = y

Number of geometrical isomers = z

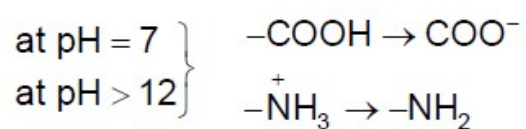
Report your answer as $x + y - z = ?$

KEY

1	2	3	4	5	6	7	8	9	10
C	A	A	A	C	C	C	B	C	ABC
11	12	13	14	15	16	17	18	19	20
ACD	BD	ABC	ABCD	ABD	ABC	BC	BCD	ABC	ABCD
21	22	23	24	25	26	27	28	29	30
5	2	2	3	7	35	7	5	8	4
31									
4									

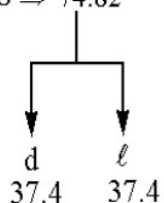
SOLUTION

1. At $\text{pH} \leq 2$ both COOH and NH_2
Remains protonated



4. O.P. $\Rightarrow -\frac{3.5}{13.9} \times 100 \Rightarrow (-) 25.18\%$

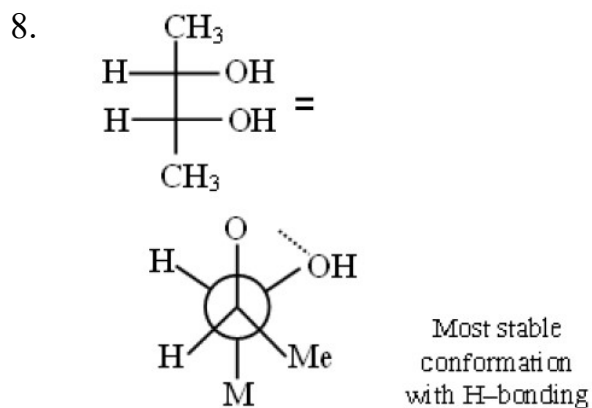
Racemic mixture $\Rightarrow 100 - 25.18 \Rightarrow 74.82$



Total d $\Rightarrow 37.4$

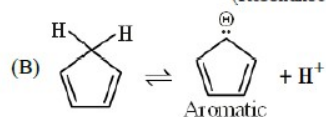
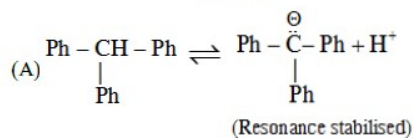
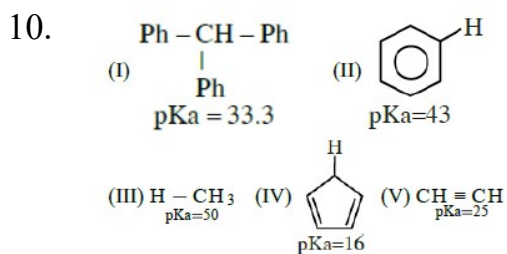
Total l $\Rightarrow 37.4 + 25.18 = 62.6$





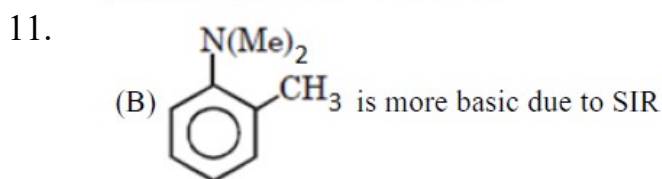
Meso butane -2,3,-diol

9. Since, the absolute configuration about the chiral carbon is (S) in each compound. Also, configuration of Cl and H about double bond carbon is also same in each compound. So, both compounds are identical.



(C) $-\text{NO}_2$ is -I group (electron withdrawing group). It increases acid strength.

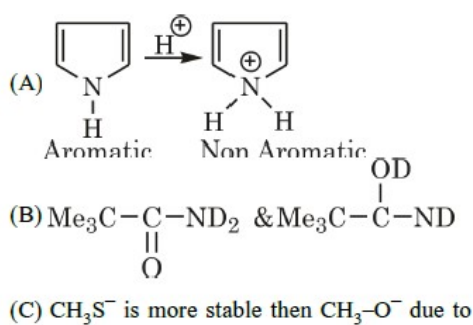
(D) Acid strength order : $\text{IV} > \text{V} > \text{I} > \text{II} > \text{III}$



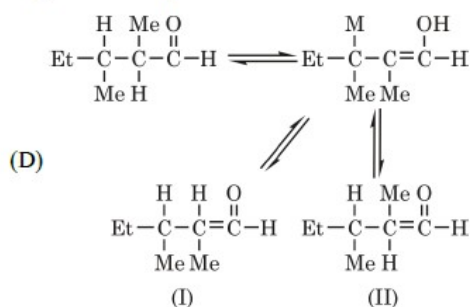
(Steric inhibition of resonance) effect

13. A lone pair of N-atom participating in resonance will be less basic

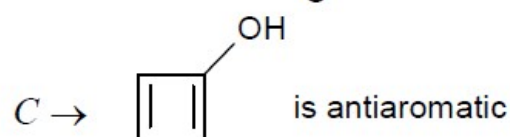
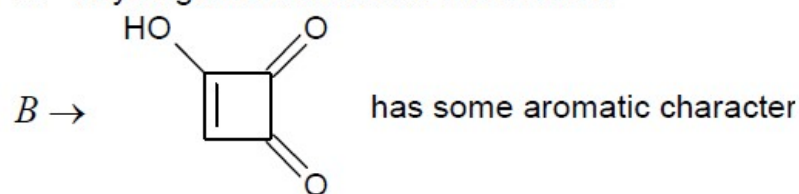
14.



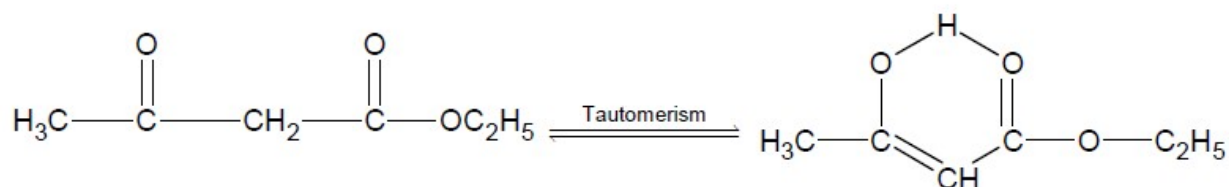
larger size of sulphur atom



15. A → hydrogen of diketone is more acidic.



16.



17.

$\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{C}(=\text{O})-\text{H}$ will show geometrical isomerism.



18.

Based on nomenclature system of geometrical isomers.

19. A – diastereomeric pair.
B – diastereomeric pair.
C – identical pair.
20. Enantiomers are also called as optical antipodes.

21. Total stereoisomers = 2^{n-1}
 $= 2^{3-1}$ ($n = 3$)
 $= 4$

Optically inactive = $2^{\frac{n-1}{2}}$
 $= 2$

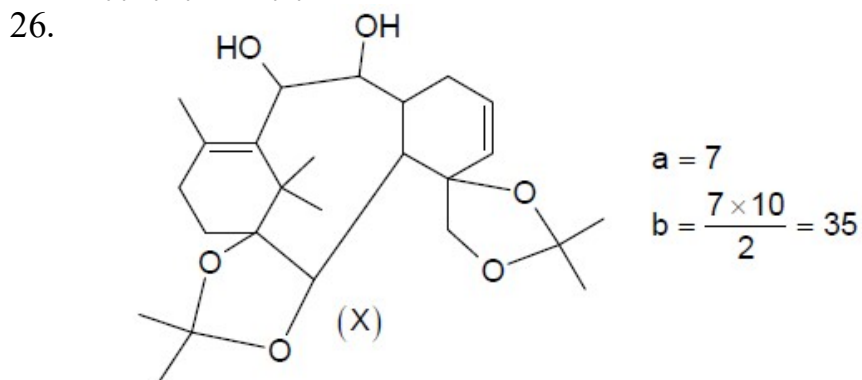
Optical active = $4 - 2 = 2$
 Number of diastereomeric pair = 5

22. Total stereoisomers = 2^{n-1}
 $= 2^{3-1}$ ($n = 3$)
 $= 4$

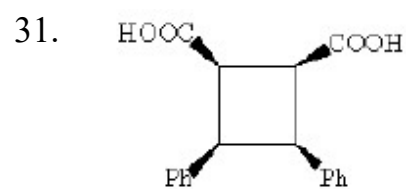
Optically inactive = $2^{\frac{n-1}{2}}$
 $= 2$

Optical active = $4 - 2 = 2$
 Number of diastereomeric pair = 5

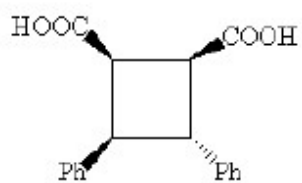
24. (i), (iii) and (v) are not aromatic.



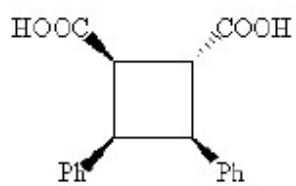
27. Number of hyperconjugative structure = number of alpha Hydrogen.



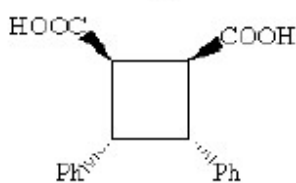
Meso



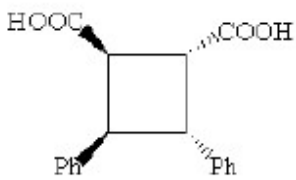
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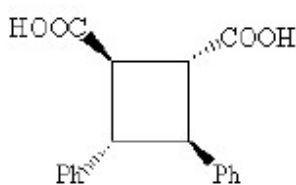
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Meso



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